THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 1

Chapter 7: **Work, Energy and Simple Machines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Work = 1

(a) force + displacement **(b)** force/displacement

(c) mass x speed **(d)** force x displacement

2. A fixed pulley: 1

(a) has MA of 2 **(b)** creates work

(c) changes the direction of force **(d)** reduces the load

3. A force of 5 N moves a body 4 m. Work: 1

(a) 0.8 J **(b)** 1.25 J **(c)** 20 J **(d)** 9 J

4. Unit of work: 1

(a) metre **(b)** newton **(c)** joule **(d)** watt

5. The fulcrum is the: 1

(a) force **(b)** effort **(c)** pivot of a lever **(d)** load

6. Work done in 10 s is 200 J. Power: 1

(a) 210 W **(b)** 20 W **(c)** 0.05 W **(d)** 2000 W

7. Unit of power: 1

(a) newton **(b)** joule **(c)** watt **(d)** pascal

8. Scissors are a: 1

(a) first class lever **(b)** third class lever

(c) pulley **(d)** second class lever

9. A 2 kg body at 5 m height (g = 10). PE: 1

(a) 10 J **(b)** 100 J **(c)** 50 J **(d)** 20 J

10. Power = 1

(a) work x time **(b)** work/time **(c)** energy x time **(d)** force/time

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Define work. 2

12. State the law of conservation of energy. 2

13. What is a lever? 2

14. Define power. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Give three examples of bodies having KE. 3

16. Find work: 20 N through 5 m. 3

17. State the energy change in a falling stone. 3

OR

Give three examples of bodies having PE. 3

18. Name the three classes of levers with an example each. 3

OR

State three differences between KE and PE. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain work and give the conditions for work to be done. 5

OR

Explain the inclined plane. 5

20. Derive KE = (1/2) m v squared. 5

OR

Explain the law of conservation of energy with an example. 5

END OF QUESTION PAPER — Total: 40 Marks

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